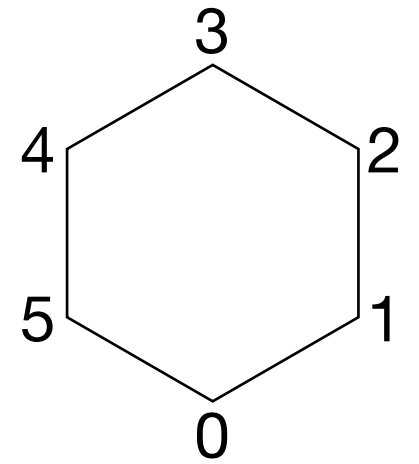
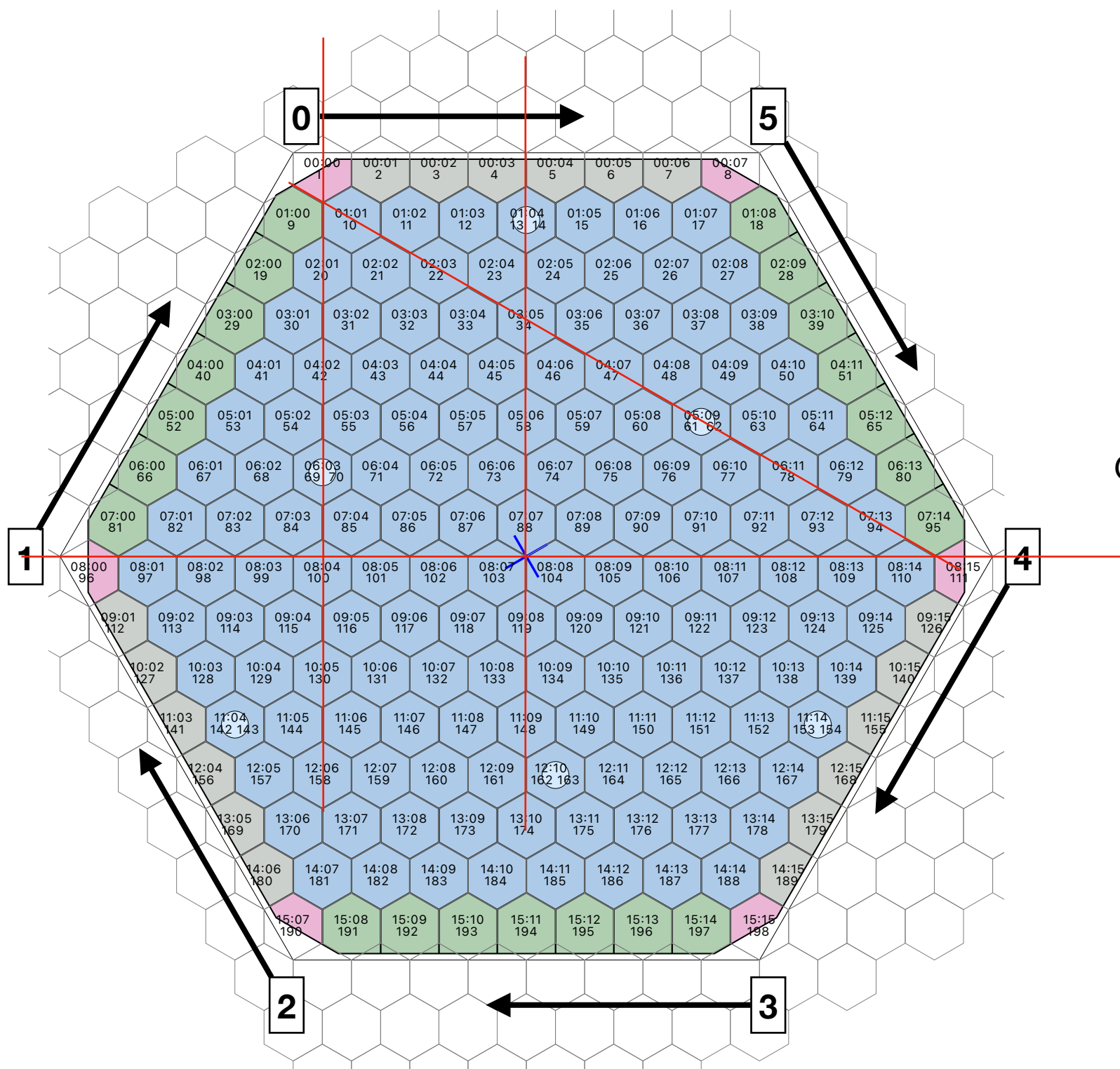
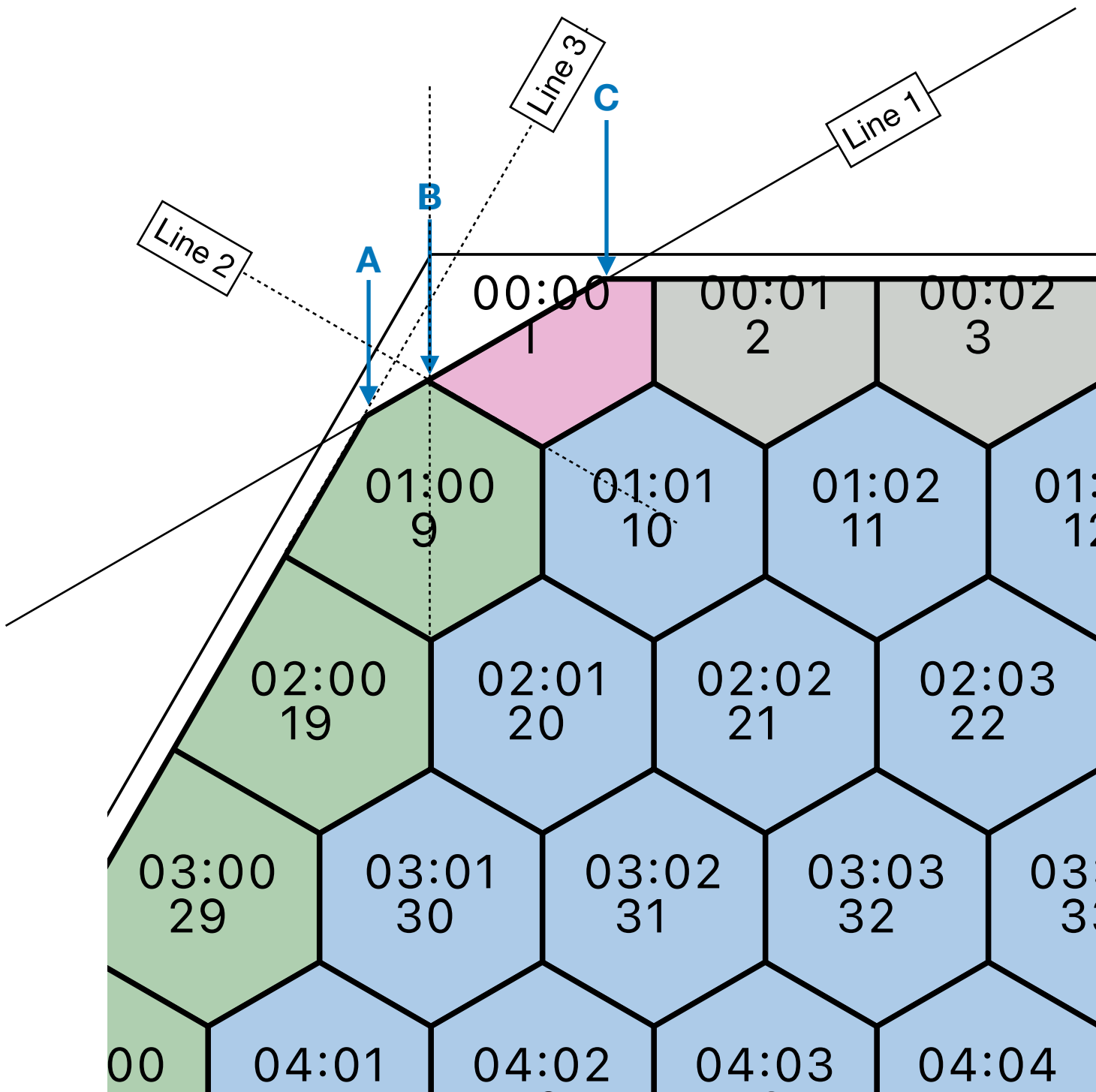




Grid cell corner numbering

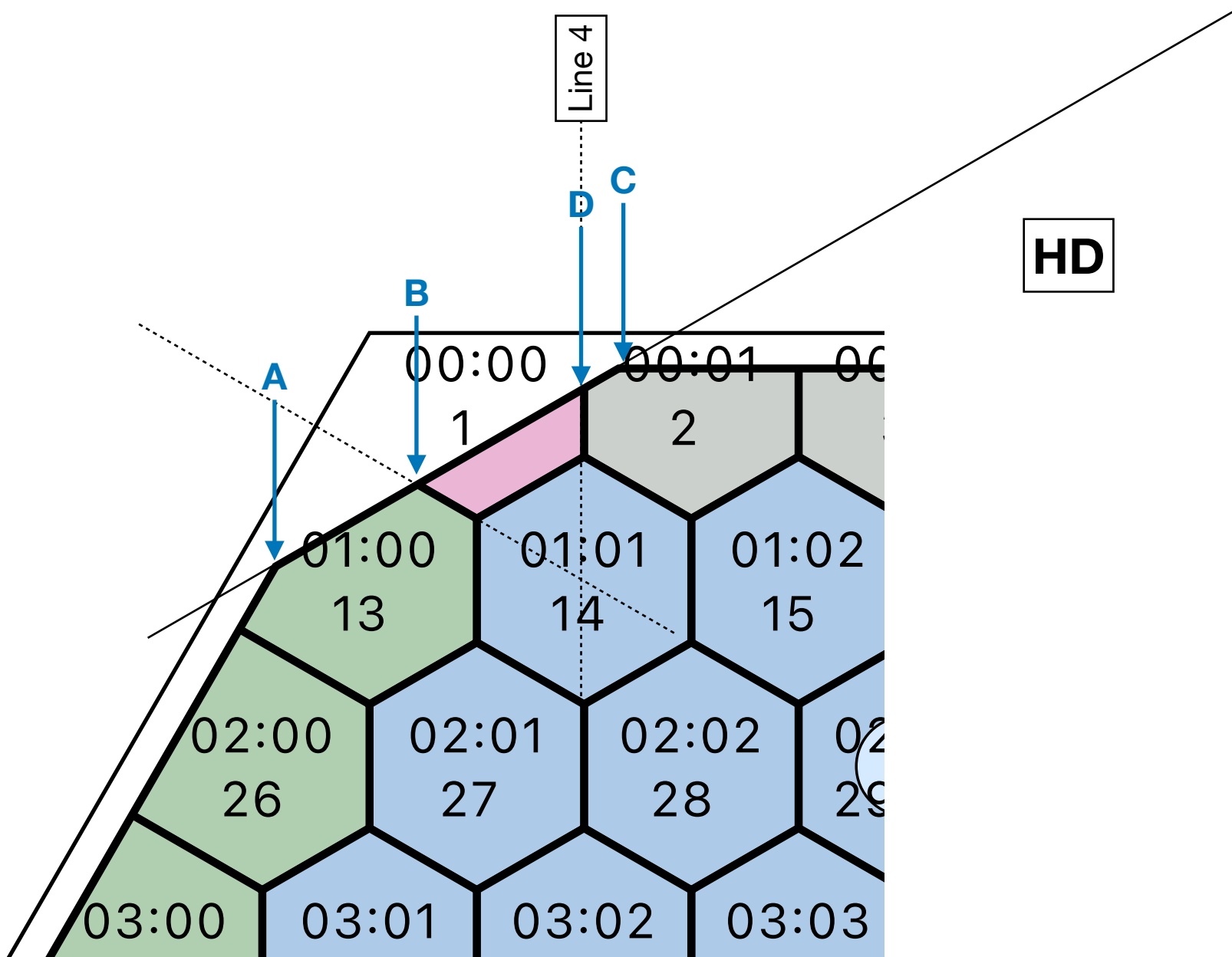


LD

Notes

$(x_c, y_c) = (-38.981, 82.385)$
 Zoltan C = $(-38.979, 82.387)$

Taking B from the vertical
 gives a point 240 μ m above
 gridpoint 5 of the corner cell



Workings for line intersections

Line 1:

$$y = mx + c \Rightarrow c = y - mx; \text{ where } m_1 = 1/\sqrt{3}$$

Point is $(x, y) = (r/2, r\sqrt{3}/2)$; where $r = \text{waferSide} - \text{mouseBite}$

$$\Rightarrow c_1 = r\sqrt{3}/2 - (1/\sqrt{3})r/2 = (r/2)(\sqrt{3} - 1/\sqrt{3})$$

Line 2:

$$y = mx + c \Rightarrow c = y - mx; \text{ where } m_2 = -1/\sqrt{3}$$

Point is $(x, y) = (-\text{width}*(n-1)/2, (\text{waferFtof}-3*\text{side})/2)$; where n is characteristic number (LD=8, HD=12)

$$\Rightarrow c_2 = (\text{waferFtof}-3*\text{side})/2 - \text{width}/\sqrt{3}(n-1)/2$$

Line 3:

$$y = mx + c \Rightarrow c = y - mx; \text{ where } m_3 = \sqrt{3}$$

Point is $(x, y) = (-ar, 0)$; where ar is active radius = $\text{activeWidth}/\sqrt{3}$

$$\Rightarrow c_3 = \text{activeWidth}$$

Point A:

intersection of Lines 1 and 3

$$m_1x + c_1 = m_3x + c_3$$

$$\Rightarrow x_b = (c_3 - c_1)/(m_1 - m_3)$$

$$y_b = m_1 x_b + c_1$$

Point B:

intersection of Lines 2 and 1

$$m_3x + c_3 = m_2x + c_2$$

$$\Rightarrow x_a = (c_2 - c_3)/(m_3 - m_2)$$

$$y_a = m_3 x_a + c_3$$

Point C:

$y = a$; where a = active half-width

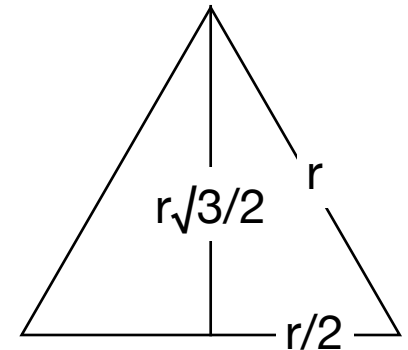
$$\Rightarrow (x_c, y_c) = ((a - c_1)/m_1, a)$$

Point D:

(needed for HD only)

$$x_d = (n/2 - 1)*\text{width}$$

$$\Rightarrow y_d = m_1 x_d + c_1$$

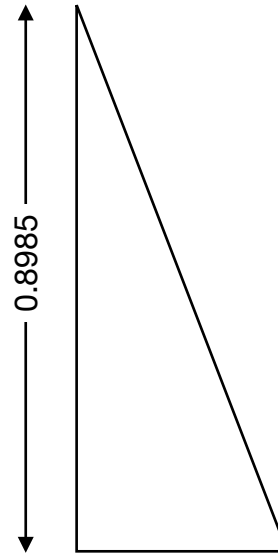


physical width, $p = 166.57\text{mm}$ (approximately)

active width (from LDzoltanA) $= 2 \times 82.3869 = 164.7738$

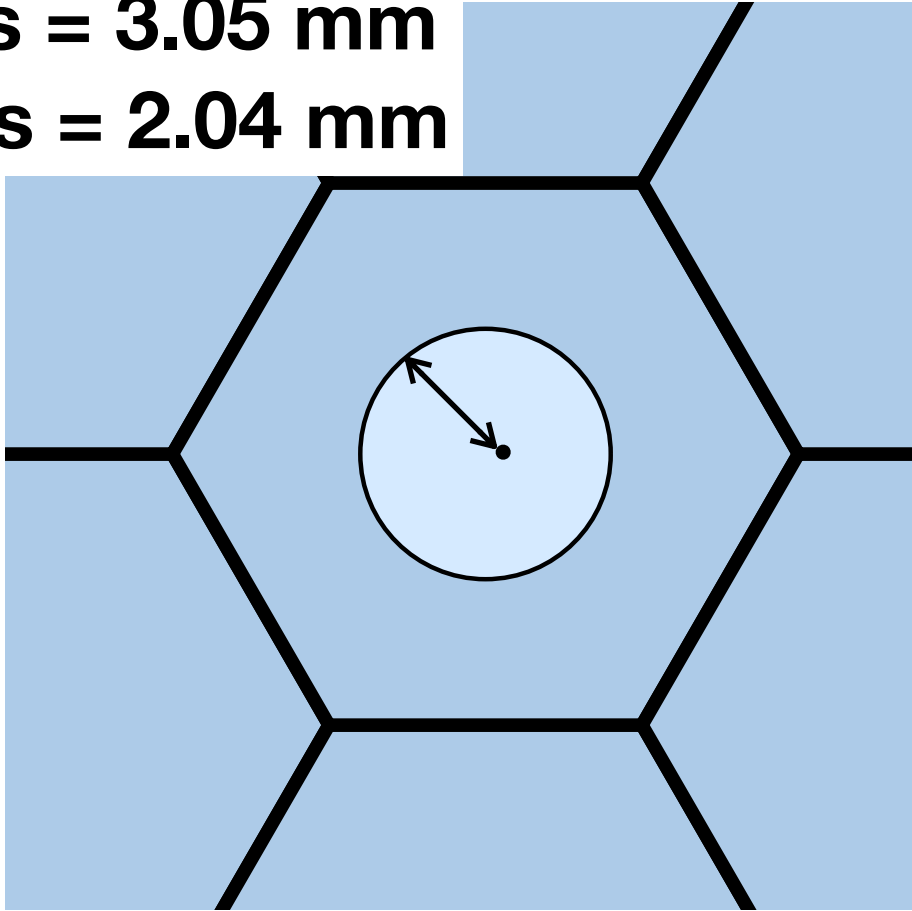
LDzoltanC $= (x_1, y_1)$

To get the physical wafer equivalent C point we can do the intersection of the line from (0,0) with $y = 166.57/2$
or simply scale the guard ring 0.8985 by x_1/y_1 to get the Δx



LD: radius = 3.05 mm

HD: radius = 2.04 mm



The centroid of a non-self-intersecting closed **polygon** defined by n vertices (x_0, y_0) , (x_1, y_1) , $\dots$, (x_{n-1}, y_{n-1}) , is the point (C_x, C_y) ,^[22] where

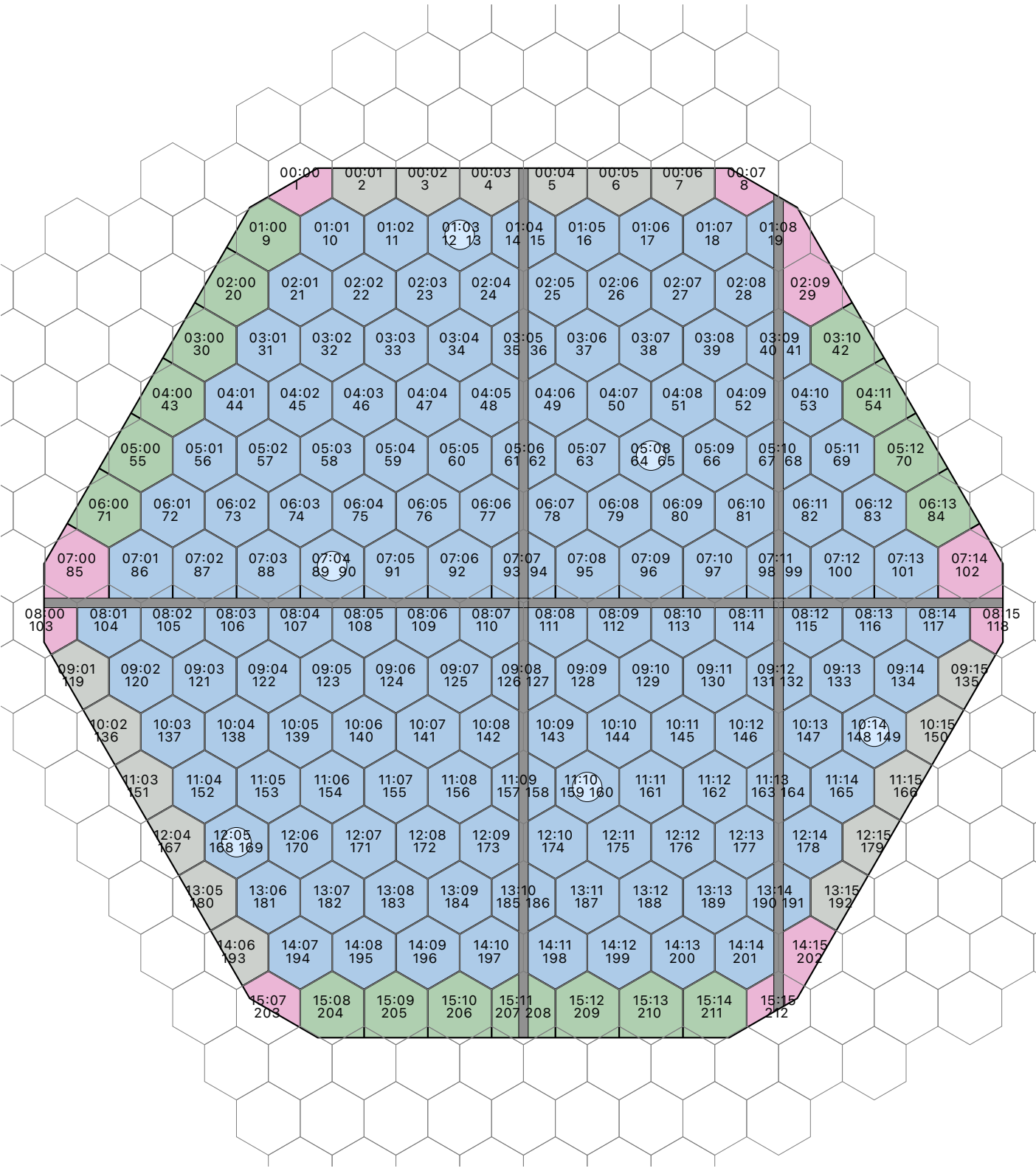
$$C_x = \frac{1}{6A} \sum_{i=0}^{n-1} (x_i + x_{i+1})(x_i y_{i+1} - x_{i+1} y_i),$$

and

$$C_y = \frac{1}{6A} \sum_{i=0}^{n-1} (y_i + y_{i+1})(x_i y_{i+1} - x_{i+1} y_i),$$

and where A is the polygon's signed area,^[22] as described by the **shoelace formula**:

$$A = \frac{1}{2} \sum_{i=0}^{n-1} (x_i y_{i+1} - x_{i+1} y_i).$$



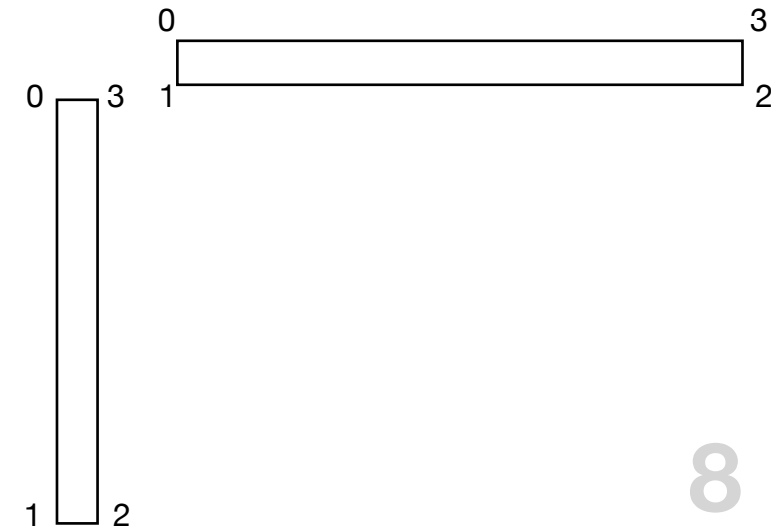
Algorithm

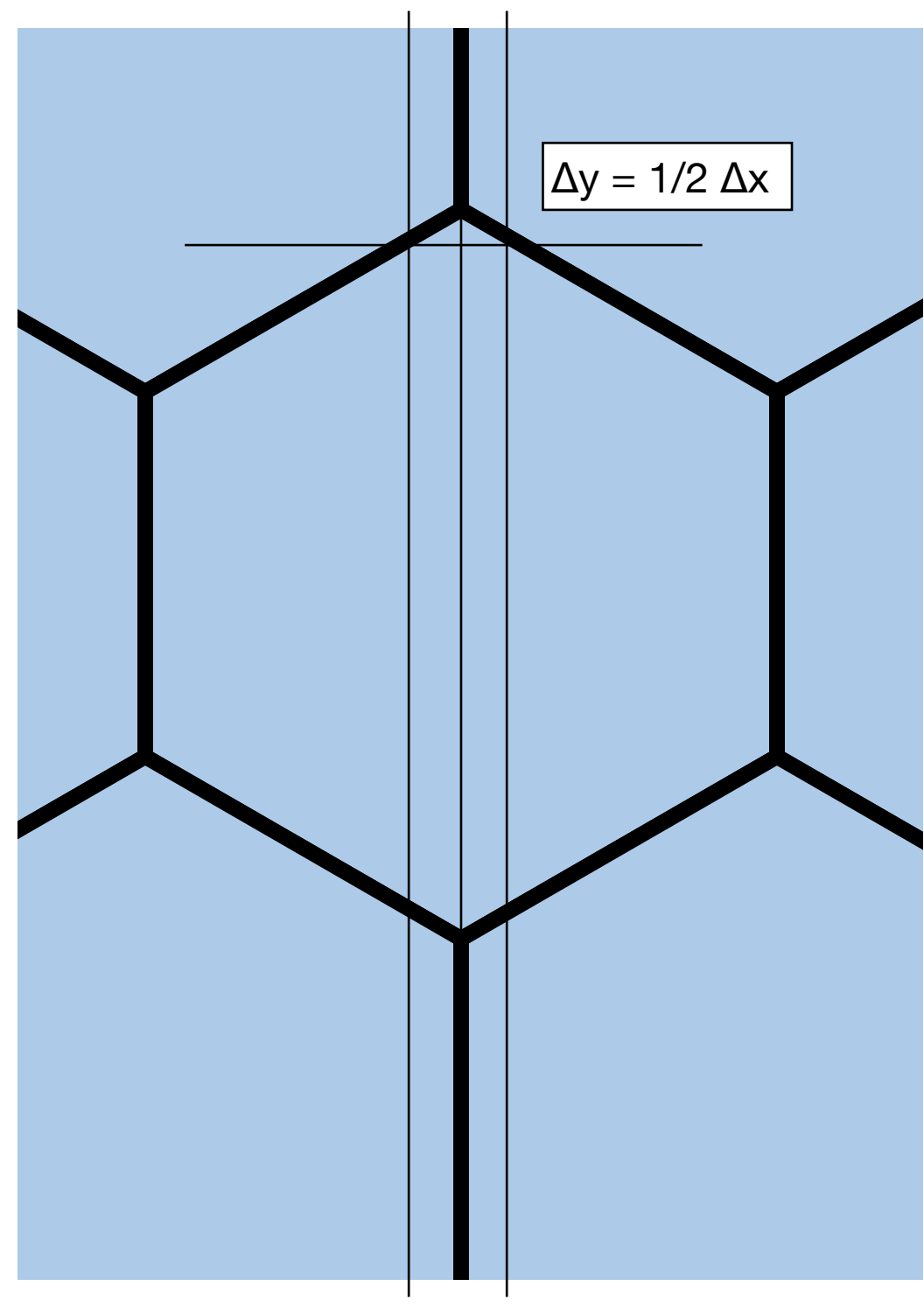
Loop over grid numbers (cellsArray)
 Make new cells and put them into a new array
 Special vertical at two places per row [1]
 depending on iu either adjust the cellCorner
 or split (for split: make two cells)
 Two special horizontal rows [1]

Deal with the special and ultra-special corners
 for LD after the loop.
 Then add calibration cells and set hardware
 numbers.

[1] If the loop is made using the by row loops
 used for the Wholes then the count value of
 special verticals and special horizontals can be
 computed OK.

New variables for StructuredCell:
 BOOL split



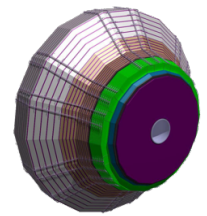


The diagram shows a light blue background with a black hexagonal lattice. A thick vertical black line runs through the center. A horizontal black line intersects the vertical line at a vertex of the hexagonal lattice. A thin vertical line is positioned to the right of the thick vertical line, passing through the same horizontal line. A small white box with a black border is located in the upper right quadrant, containing the equation $\Delta y = 1/2 \Delta x$.

$$\Delta y = 1/2 \Delta x$$

Mouse-bite size

Chris Seez
HGCAL GEANT geometry
May 2024



The mouse-bite

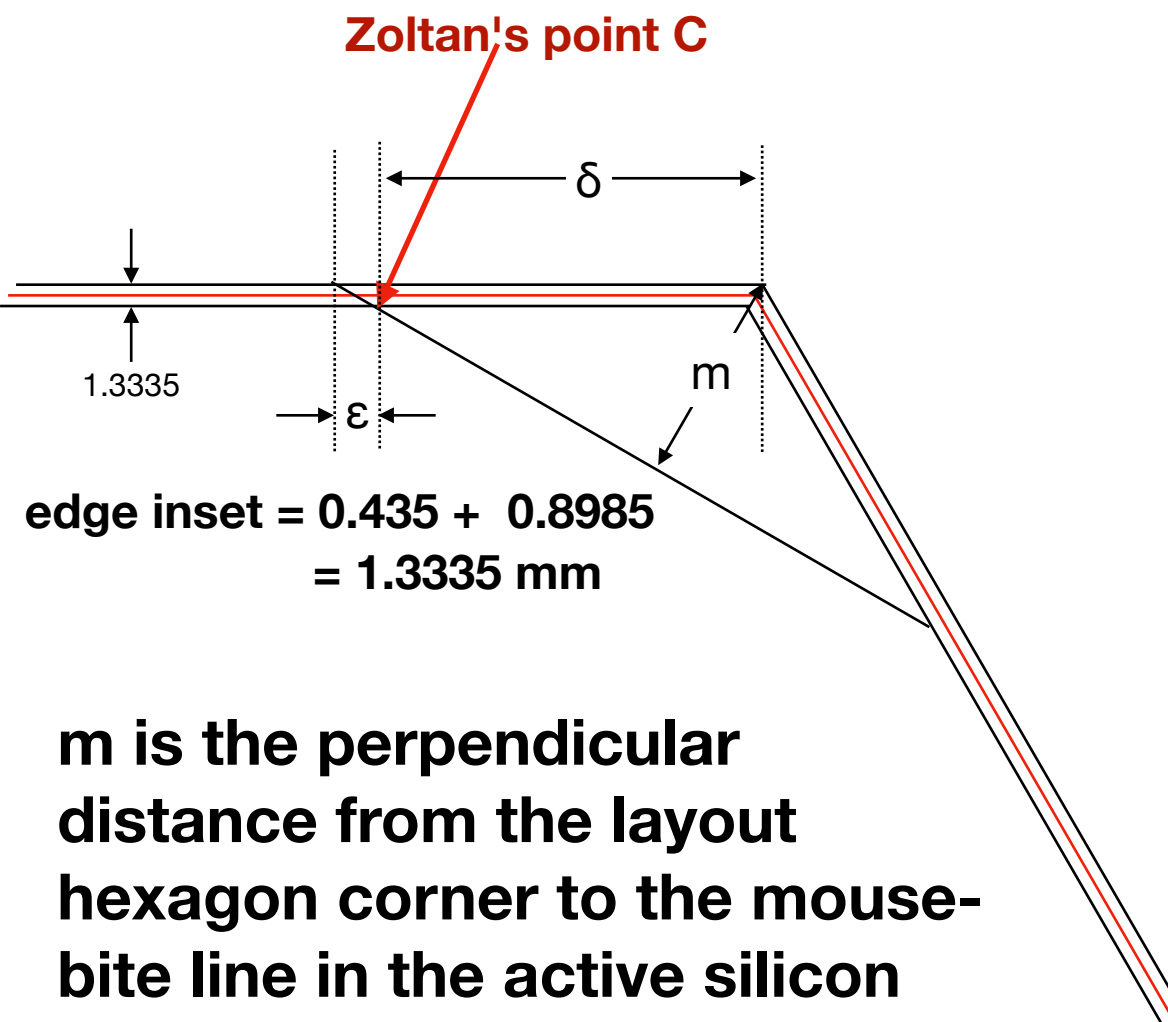


Diagram shows one corner of wafer hexagon

Three hexagons:

Layout hexagon: 167.441 mm

Physical wafer: 166.571 mm

Sensitive cells: 164.774 mm

The x-coordinate of Zoltan's point C is 38.9787 mm (wrt origin at wafer centre)

Using the 30/60° right-angle triangles:

$$\delta = 0.5 \times (167.441/\sqrt{3}) - 38.9787$$

$$= 9.357 \text{ mm}$$

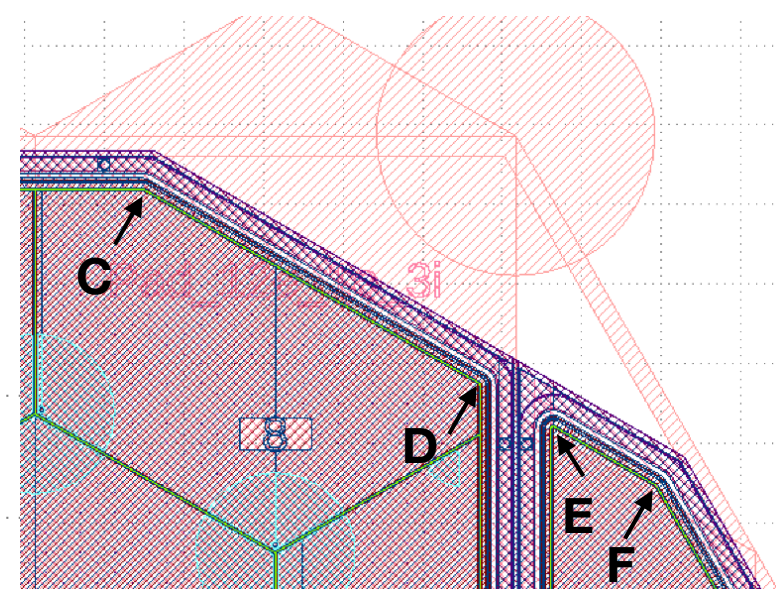
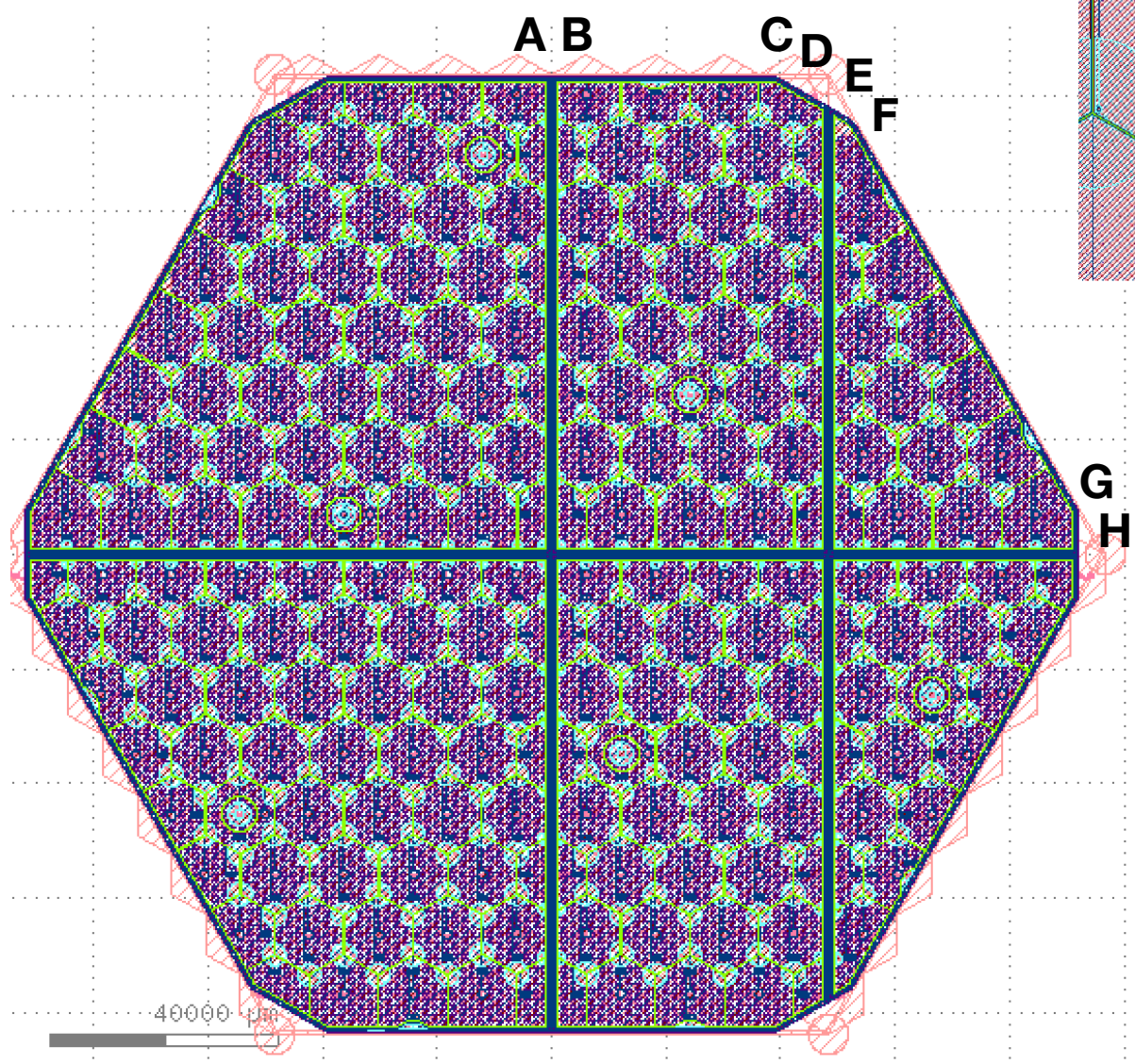
$$\epsilon = 1.334 \times \sqrt{3}$$

$$= 2.310 \text{ mm}$$

$$\text{So } m = (9.357 + 2.310)/2$$

$$= 5.834 \text{ mm}$$

x,y coordinates in μm with respect to wafer centre
from Zoltan Gesce



	x	y
A	-898.5	82386.9
B	898.5	82386.9
C	38978.7	82386.9
D	47437.5	77503.2
E	49234.5	76465.7
F	51859.8	74950
G	90838.5	7436.9
H	90838.5	898.5

diceVert2LD = 48.336

diceVert1HD = -28.196

diceVert2HD = 28.196

diceHorzHD = 16.279

A	27297.5	82386.9
B	29094.5	82386.9
C	38978.7	82386.9
D	51859.8	74950
E	85214.8	17177.5
F	86252.3	15380.5
G	90838.5	7436.9

